

A two-point counterexample to metric-frame existence

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Source and Target

The source paper is K. Mahesh Krishna’s arXiv:2202.13697 paper [1]. Its title is *Metric, Schauder and Operator-Valued Frames*. In Section 2.1, after defining metric frames, the source states on PDF page 68 that, after the definition of metric frame, “the first question which comes is the existence.” It then explains that the analogous Banach-frame existence theorem is known for separable Banach spaces, while the metric-space setting is not known in that form.

We now define $S : \mathbb{K}^m \ni (x_1, \dots, x_m) \mapsto \sum_{j=1}^m x_j \in \mathbb{K}$. Then $(\{f_n\}_n, S)$ is a metric frame for $\mathbb{K}$. Note that the operator S is linear.

After the definition of metric frame, the first question which comes is the existence. In Theorem 1.3.13 it was proved that every separable Banach space admits a Banach frame. Even though this result is not known in metric space settings, two results are derived one is close to the definition of metric frame and another gives existence under certain assumptions. To do this we prove a result which we derive from Mc-Shane extension theorem.

Theorem 2.1.15. (Mc-Shane extension theorem) (cf. [Weaver \(2018\)](#)) Let $\mathcal{M}$ be a metric space and $\mathcal{M}_0$ be a nonempty subset of $\mathcal{M}$. If $f_0 : \mathcal{M}_0 \rightarrow \mathbb{R}$ is Lipschitz, then there exists a Lipschitz function $f : \mathcal{M} \rightarrow \mathbb{R}$ such that $f|_{\mathcal{M}_0} = f_0$ and $\text{Lip}(f) = \text{Lip}(f_0)$.

Using Theorem 2.1.15 we derive the following.

Corollary 2.1.16. If $(\mathcal{M}, 0)$ is a pointed metric space, then for every $x \in \mathcal{M}$, there exists a Lipschitz function $f : \mathcal{M} \rightarrow \mathbb{R}$ such that $f(x) = d(x, 0)$, $f(0) = 0$ and $\text{Lip}(f) = 1$.

Proof. Case (i) : $x \neq 0$. Define $\mathcal{M}_0 := \{0, x\}$ and $f_0(0) = 0$, $f_0(x) = d(x, 0)$. Then $|f_0(x) - f_0(0)| = d(x, 0)$ and hence f_0 is Lipschitz. Application of Theorem 2.1.15 now completes the proof.

Case (ii) : $x = 0$. Take a non-zero point $y \in \mathcal{M}$. We use the argument in case (i) by replacing y in the place of x . $\square$

Theorem 2.1.17. Let $\mathcal{M}$ be a separable metric space. Then there exist a BK-space $\mathcal{M}_d$, a sequence $\{f_n\}_n$ in $\text{Lip}_0(\mathcal{M}, \mathbb{R})$ and a function $S : \mathcal{M}_d \rightarrow \mathcal{M}$ such that

- (i) $\{f_n(x)\}_n \in \mathcal{M}_d$, for each $x \in \mathcal{M}$,
- (ii) $\|\{f_n(x) - f_n(y)\}_n\|_{\mathcal{M}_d} \leq d(x, y)$, $\forall x, y \in \mathcal{M}$,
- (iii) $S(\{f_n(x)\}_n) = x$, for each $x \in \mathcal{M}$.

Proof. Let $\{x_n\}_n$ be a dense set in $\mathcal{M}$. Then for each $n \in \mathbb{N}$, from Corollary 2.1.16 there exists a Lipschitz function $f_n : \mathcal{M} \rightarrow \mathbb{R}$ such that $f_n(x_n) = d(x_n, 0)$, $f_n(0) = 0$ and

Definitions

We use the source's Definition 2.1.10. A metric frame for a metric space $\mathcal{M}$, with respect to a BK-space $\mathcal{M}_d$, consists of Lipschitz functions $f_n : \mathcal{M} \rightarrow \mathbb{K}$ and a Lipschitz reconstruction operator $S : \mathcal{M}_d \rightarrow \mathcal{M}$ such that the analysis map

$$\theta_f(x) = (f_n(x))_{n \geq 1} \in \mathcal{M}_d$$

is bi-Lipschitz onto its image and

$$S(\theta_f(x)) = x, \quad x \in \mathcal{M}.$$

A BK-space is, in particular, a normed linear space and hence is path connected.

Proof Intuition

The reconstruction operator in a metric frame is stronger than a formal coordinate inverse: it is a Lipschitz map from the whole Banach sequence space $\mathcal{M}_d$ onto $\mathcal{M}$. Since a Banach space is linearly path connected, composing the line segment between two analysis vectors with S gives a path in $\mathcal{M}$. Thus metric-frame reconstruction forces the underlying metric space to be path connected. A two-point metric space is separable but not path connected, so it cannot admit such a frame.

Theorem 1. *If a metric space $\mathcal{M}$ admits a metric frame in the sense of Definition 2.1.10 of [1], then $\mathcal{M}$ is Lipschitz path connected.*

Proof. Let $(\{f_n\}_{n \geq 1}, S)$ be a metric frame for $\mathcal{M}$ with coefficient BK-space $\mathcal{M}_d$. Fix $x, y \in \mathcal{M}$. Put $u = \theta_f(x)$ and $v = \theta_f(y)$. Since $\mathcal{M}_d$ is a normed linear space, the segment

$$\sigma(t) = (1 - t)u + tv, \quad 0 \leq t \leq 1,$$

is a Lipschitz path in $\mathcal{M}_d$. The composition $\gamma = S \circ \sigma : [0, 1] \rightarrow \mathcal{M}$ is Lipschitz because S is Lipschitz. Moreover,

$$\gamma(0) = S(\theta_f(x)) = x, \quad \gamma(1) = S(\theta_f(y)) = y.$$

Thus every two points of $\mathcal{M}$ are joined by a Lipschitz path. □

Corollary 1. *The two-point metric space $\mathcal{M} = \{0, 1\}$, with $d(0, 1) = 1$, is a separable metric space which admits no metric frame in the sense of Definition 2.1.10 of [1].*

Proof. The space $\{0, 1\}$ is separable. It is not path connected: the continuous image of the connected interval $[0, 1]$ into the discrete two-point metric space must be a singleton. This contradicts the theorem if a metric frame existed. □

Verification Notes

The argument uses only the Lipschitz reconstruction clause in the definition of metric frame. It does not use the lower or upper frame inequalities except to ensure that the analysis vectors lie in the BK-space.

This counterexample does not contradict the source's Theorem 2.1.17. That theorem gives Lipschitz coordinate functions and a reconstruction function, but its statement does not require the reconstruction function to be Lipschitz. The obstruction above applies precisely to the full metric-frame definition, where $S : \mathcal{M}_d \rightarrow \mathcal{M}$ is required to be Lipschitz on the whole BK-space.

No computational checks were needed. The source PDF and evidence crop are included in this packet.

Novelty and Literature Check

The run indexes were searched before promotion for arXiv:2202.13697 and for core phrases including **metric frame**, **Lipschitz retract**, and **path connected**. No prior packet or attempt for this target was found. Web and arXiv searches on 2026-06-28 for the exact source title and phrases such as **metric frame separable metric**, **metric frame Lipschitz retract**, and **metric frame path connected** found the source paper, the related arXiv:2011.01870 paper, and adjacent Lipschitz-frame work, but no duplicate of this path-connectedness obstruction.

Human Review Recommendation

Check the source definition of metric frame, especially that the reconstruction operator S is required to be Lipschitz on all of $\mathcal{M}_d$. Under that definition, the proof above is immediate and the two-point space refutes the broad existence question for arbitrary separable metric spaces.

References

- [1] K. Mahesh Krishna. *Metric, Schauder and Operator-Valued Frames*. arXiv:2202.13697, 2022.